

Take-Home Assignment

Welcome, and thank you for your interest in the role at **durchblicker**. This assignment is your chance to show how you work on a realistic data scenario.

Context

At durchblicker, data flows in from several sources (CRM exports, tracking, manual CSVs). These are often inconsistently formatted, contain duplicates, and KPIs are calculated differently depending on the source. Your task is to build a clean, documented data model from such raw data.

We provide you with two CSV files: **leads.csv** (one row per lead interaction with a comparison calculator) and **conversions.csv** (one row per closed contract, linked to a lead). They simulate a lead-and-conversion dataset.

Datasets

leads.csv

- lead_id, email, vertical (auto/haushalt/leben), source (google/newsletter/direct), created_at, region

conversions.csv

- conversion_id, lead_id, email, premium (annual net premium), signed_date, status (active/cancelled/pending)

Ground Rules

You may use AI tools as much as you like. We're not assessing whether you use them, but whether you understand, question and can justify the results. In the interview we'll discuss your solution and your decisions.

We'd rather see less that's well thought through than everything done halfway. Briefly document your assumptions and open questions, that's explicitly part of the assessment.

Tasks

Task 1 – Data Profiling & Data Quality

Before you model anything: get yourself, and us, an overview of the data quality across both files.

Deliver a short profiling: row counts, missing values, duplicates, suspicious values, value ranges. From that, produce a list of concrete data quality issues, and for each one say how critical you consider it and how you'd handle it.

Task 2 – Lead-to-Conversion Model

Build a model that links leads to their conversions, so a business team can see which leads converted and which didn't. The result should be a table people can actually work with.

Make a justified decision on how to handle conversions that can't be matched to a lead, and leads that never converted.

Task 3 – KPI & Customer-Level Aggregation

Build aggregated models that answer questions the business actually asks. At minimum:

- Per (vertical, source, month): number of leads, number of conversions, and conversion rate.
- A customer-level view (one row per customer): number of leads, number of conversions, total active premium, and whether the customer currently holds an active contract.

Make your definitions explicit, especially “what counts as a conversion” and “what is a customer”.

Task 4 – Architecture & Automation (conceptual + some code)

Sketch out how you'd turn these one-off scripts into a maintainable, recurring pipeline. We don't expect runnable infrastructure here, but clear thinking plus some illustrative code.

Address the following:

- Layer structure (raw/staging/mart or similar).
- Where data quality checks sit (e.g. dbt tests / Dataform assertions).
- How often this should run and what happens on failure.
- How you ensure traceability and documentation.

Show concrete example code for one of the tests or assertions.

Submission

- Send us your code along with a short write-up of your assumptions and open questions.
- Language and tools are up to you (SQL, Python, dbt/Dataform ...), pick whatever lets you best show what you can do.
- In the follow-up conversation we'll walk through your solution together.

We're looking forward to your solution. Enjoy!